Powershell:
Blue Screen of Death but not Dead



What is the Shell of Power?

Pros:

- Said to replace CMD & Batch
- Can be run on other OS
  - Why? IDK, Linux Powershell = Fun

Can automated anything, but not well

Get to learn English vocabulary-

Verb = Action Word

Noun = Thing doing Action

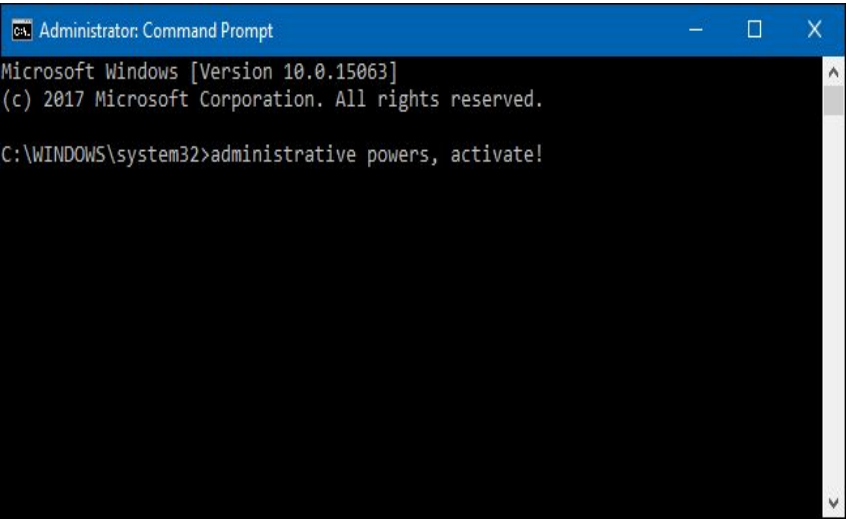
It is not Java! (No more Karel)

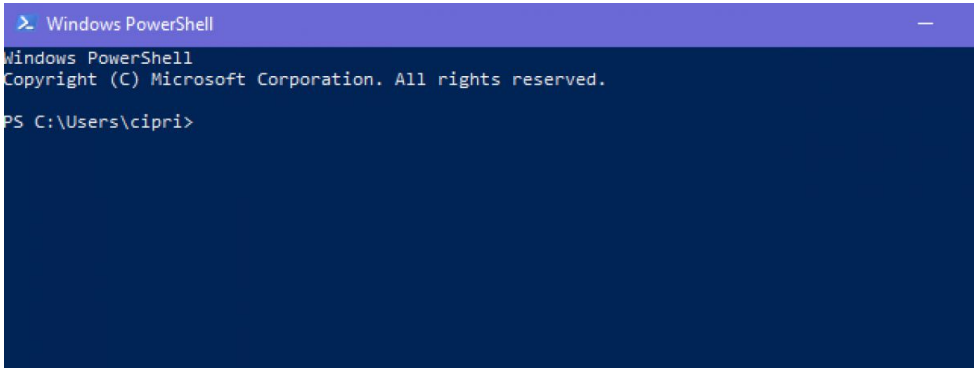
Cons:

Scratch is a superior language



Better Command Prompt





Powershell is blue, therefore it is better.

Worse Programming Language

Powershell uses Verb-Noun structure. A verb describes what it does, a noun describes what it is. Get-EventLog retrieves the event log.

Java Syntax = verbNoun = Troy.getName()
Powershell Syntax = Verb-Noun = Get-Name Troy




```
PS C:\> Get-Verb | Sort-Object -Property Verb

Verb        Group
----        -----
Add         Common
Approve     Lifecycle
Assert      Lifecycle
Backup      Data
Block       Security
Checkpoint  Data
Clear       Common
Close       Common
Compare     Data
Complete    Lifecycle
Compress    Data
```

Note on Powershell

Powershell documents its commands where each word has its first letter capitalized

You do not have to capitalize letters when typing in a powershell command - it’s simply just standardization

Powershell <3's CMD



Powershell has the ability to call Command Prompt executables that are included in the `%path%` variable

However it will prioritize its own aliases first

For example: in powershell you can still use "`cd`" to change directory

However cd is actually an alias for the "`set-location`" command and not the cmd executable

```
PS C:\Users\800018542> get-alias cd

CommandType     Name
-----------     ----
Alias           cd -> Set-Location
```

B-But my [insert useless command]!

- Sometimes, you really wanna use a classic CMD command
- To make sure you pull this off type "`cmd /c`" before the command
  - For instance if you want to use the classic copy command
  - `cmd /c copy [file] [destination]`



Powershell includes *some* linux

- *Some* Linux commands are aliased
- You can use “ls” instead of “dir” to list the current directory

More on aliases

See what an alias refers to:

`get-alias [alias]`

If you don't specify it will list all commands

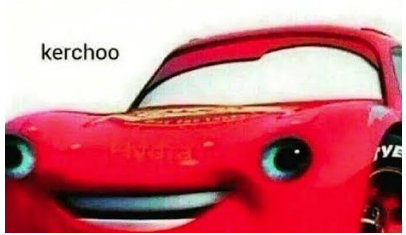
```
PS C:\Users\800018542> get-alias

CommandType     Name
-----------     ----
Alias           % -> ForEach-Object
Alias           ? -> Where-Object
Alias           ac -> Add-Content
Alias           asnp -> Add-PSSnapin
Alias           cat -> Get-Content
Alias           cd -> Set-Location
Alias           CFS -> ConvertFrom-String
Alias           chdir -> Set-Location
```

Check if a command has aliases:

`get-alias -definition [command]`

Useful to *become speed*.



Drives



There are many drives in powershell

Use Get-PsDrive to see them all

```
PS C:\Users\800018542> get-psdrive

Name           Used (GB)     Free (G
----           ---------     -------
Alias
C                 94.00           3.
Cert
D                  6.70           0.
E
Env
Function
HKCU
HKLM
M              -2020.15        2020.
Variable
W                 95.28         730.
WSMan
```

As you can see Alias: is a drive, along with Function: and C:

Registry is HKLM: and HKCU:, but aren't in this presentation

Env: refers to CMD environmental variables

A common output manipulator command

- Select-Object
  - Takes the output of another command and allows you to do stuff like filtering or listing properties

```
Get-ADUser -Identity Test.User -Properties * | Select-Object -ExpandProperty
DisplayName,SamAccountName,Description
```

Another common output manipulator command

- **Format-Wide**, **Format-List**, **Format-Table**, and **Format-Custom**
  - Formats output of another command

```
$users = Get-ADUser -Filter * | Format-List
# users is a list of users
```

Help Plz

There's not enough powershell problems for StackOverflow to be useful…

Instead use:

- Microsoft Docs
- Get-Help



Scripts that Script the Scripting of Scripts

Powershell ISE:

- Microsoft has their own built-in scripting environment for Windows.
- (ISE = Integrated Scripting Environment)



Visual Studio Code:

- What Macrosoft wants you to use to code Powershell. It can also run python, java, C#, and C++



Autocomp - DID YOU MEAN autocomplete?

- Autocomplete is useful because when writing a script you want to use the full notation for all the commands inside your script

- try to forget the “smallest # of characters” challenges because you should know what the hell your code is calling

- NO ALIASES IN YOUR CODE
  - this means no CD, LS, CP etc
  - i suppose stuff like foreach is okay but don’t push the conventions too far

Why Power  ?

- the whole purpose of scripting in powershell rather than batch is to play around with Objects better
- string/int/data types/classes etc are all important and you want to know the difference, how to manipulate them in order to pull information easier